

Power Inputs

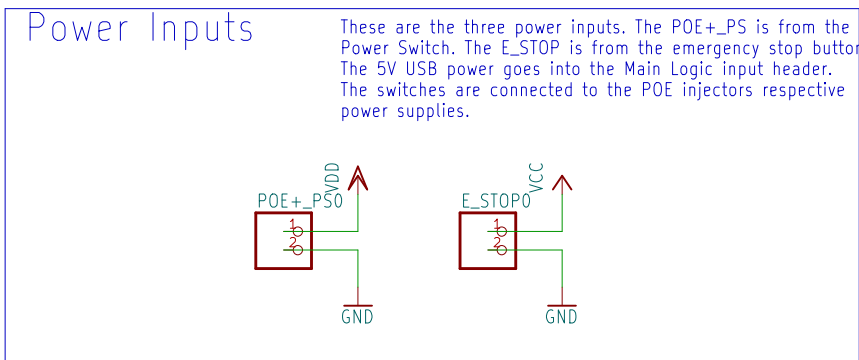
These are the three power inputs. The POE+_PS is from the Power Switch. The E_STOP is from the emergency stop button. The 5V USB power goes into the Main Logic input header. The switches are connected to the POE injectors respective power supplies.

The diagram consists of two separate circuit sections. The left section shows a red rectangular box containing a switch symbol. A green line labeled 'POE+_PS' enters the top of the box. Another green line exits the bottom of the box and connects to a green ground symbol labeled 'GND'. A third green line exits the right side of the box and connects to a green arrow pointing upwards, labeled 'VDD'. The right section is identical in structure. It shows a red rectangular box containing a switch symbol. A green line labeled 'E_STOP' enters the top of the box. Another green line exits the bottom of the box and connects to a green ground symbol labeled 'GND'. A third green line exits the right side of the box and connects to a green arrow pointing upwards, labeled 'VCC'.

Power Inputs

These are the three power inputs. The POE+_PS is from the Power Switch. The E_STOP is from the emergency stop button. The 5V USB power goes into the Main Logic input header. The switches are connected to the POE injectors respective power supplies.

The image contains two circuit diagrams. The left diagram shows a switch labeled 'POE+_PS' connected to 'VDD' and 'GND'. The right diagram shows a switch labeled 'E_STOP' connected to 'VCC' and 'GND'.



Main Logic

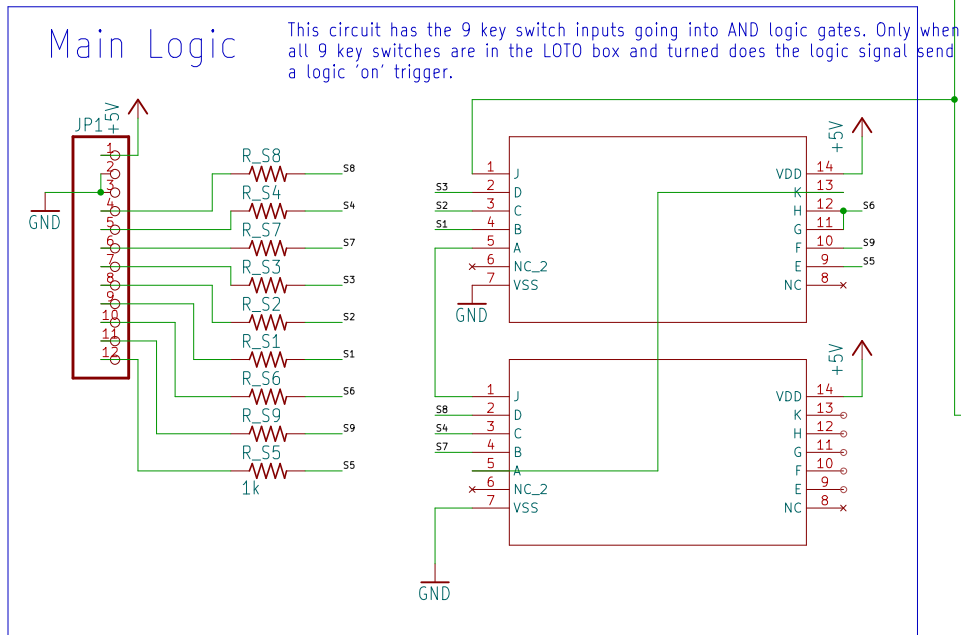
This circuit has the 9 key switches going into AND logic gates. Only when all 9 key switches are in the LOTO box and turned does the logic signal send a logic 'on' trigger.

The circuit consists of two 9-input AND gates. The first gate has inputs J (1), D (2), C (3), B (4), A (5), NC_2 (6), VSS (7), S3 (8), and S2 (9). The second gate has inputs J (1), D (2), C (3), B (4), A (5), NC_2 (6), VSS (7), S8 (8), and S4 (9). Both gates have VDD (14), K (13), H (12), G (11), F (10), E (9), and NC (8) pins. The output of the first gate is S6 and the output of the second gate is S9. A 1k resistor is connected to the output of the first gate.

Main Logic

This circuit has the 9 key switches going into AND logic gates. Only when all 9 key switches are in the LOTO box and turned does the logic signal send a logic 'on' trigger.

The circuit consists of two 9-input AND gates. The first gate has inputs J (1), D (2), C (3), B (4), A (5), NC_2 (6), VSS (7), S3 (8), S2 (9), and S1 (10). The second gate has inputs J (1), D (2), C (3), B (4), A (5), NC_2 (6), VSS (7), S8 (8), S4 (9), and S7 (10). Both gates have VDD (14), K (13), H (12), G (11), F (10), E (9), and NC (8) pins. The output of the first gate is S6 and the output of the second gate is S5. A 1k resistor is connected to the output of the first gate. A 5V supply is connected to the VDD pins of both gates.



Main Logic

This circuit is triggered by the logic output. If 'on' the power from the POE+ (power switch) will be connected, if 'off' the POE+ injector will be off.

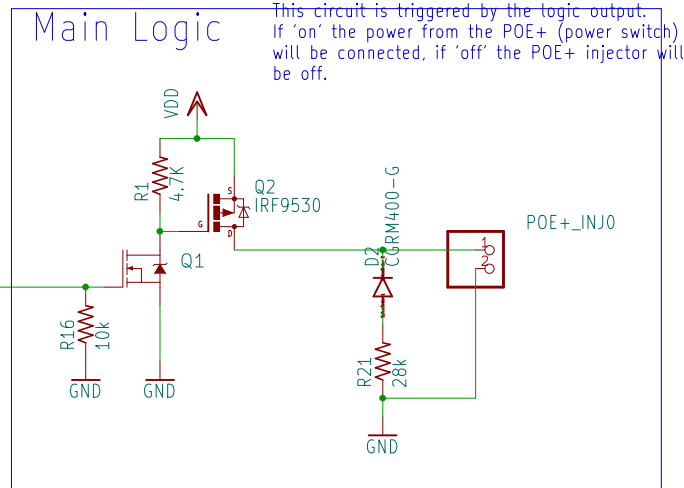
The circuit diagram illustrates the main logic for the POE injector. It features two MOSFETs, Q1 and Q2, both labeled IRF9530. Q1 is an N-channel MOSFET with its source connected to GND and its gate connected to the logic input. Its drain is connected to the gate of Q2. Q2 is a P-channel MOSFET with its source connected to GND and its gate connected to the drain of Q1. Its drain is connected to the positive terminal of a relay, represented by a rectangle with a circle inside. The relay's other terminal is connected to the POE+_INJ0 input. A 4.7k resistor (R1) is connected between VDD and the gate of Q2. A 10k resistor (R16) is connected between the logic input and GND. A 28k resistor (R21) is connected between the relay's positive terminal and GND. A diode D2 (CGRM400-G) is connected in parallel with the relay, with its cathode to the positive terminal and its anode to GND.

Main Logic

This circuit is triggered by the logic output. If 'on' the power from the POE+ (power switch) will be connected, if 'off' the POE+ injector will be off.

The circuit diagram shows the following components and connections:

- Logic Input:** Connected to the gate of MOSFET Q1 (IRF9530).
- Resistor R16:** 10k, connected between the logic input and GND.
- MOSFET Q1:** IRF9530, N-channel MOSFET. Source is connected to GND. Drain is connected to the gate of MOSFET Q2.
- Resistor R1:** 4.7k, connected between VDD and the gate of MOSFET Q2.
- MOSFET Q2:** IRF9530, N-channel MOSFET. Source is connected to GND. Drain is connected to the positive terminal of the relay (D2).
- Relay D2:** CDRW400-G, represented by a red box with a circle and a diagonal line through it. One terminal is connected to the drain of Q2, and the other terminal is connected to the POE+_INJ0 input.
- Resistor R21:** 28k, connected between the relay terminal and GND.



Main Logic

This circuit is triggered by the logic output. If 'on' the power from the POE (e-stop) will be connected, if 'off' the POE injector will be off.

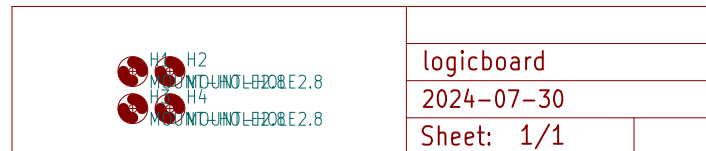
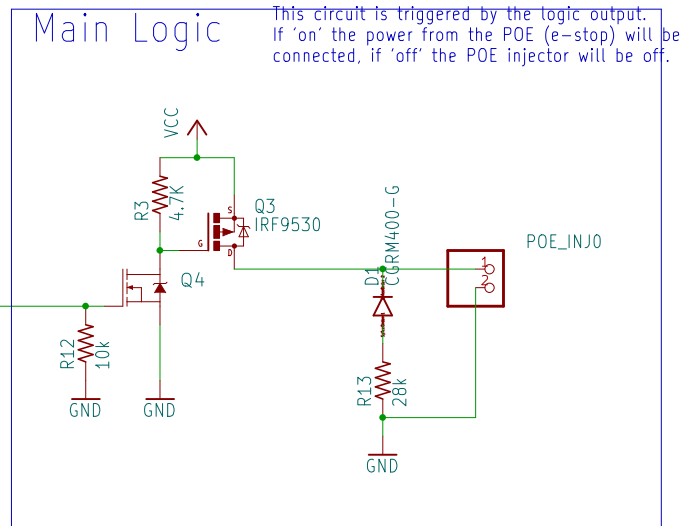
The circuit diagram illustrates the main logic for controlling the POE injector. It features a MOSFET driver stage and a relay control stage.

- Input Stage:** A green line representing the logic output is connected to the gate of MOSFET Q4 (IRF9530). A pull-up resistor R2 (10k) connects the input to VCC.
- Driver Stage:** MOSFET Q4's source is grounded, and its drain is connected to the gate of MOSFET Q3 (IRF9530). A resistor R3 (4.7k) connects the gate of Q3 to VCC.
- Relay Control Stage:** MOSFET Q3's source is grounded, and its drain is connected to the coil of a relay (represented by a rectangle with a circle inside). A diode D1 (CGRM400-G) is connected in parallel with the relay coil, with its cathode to the MOSFET drain and its anode to ground.
- Output Stage:** A resistor R13 (28k) connects the relay coil to ground. The other terminal of the relay coil is connected to the POE_INJO output.

Main Logic

This circuit is triggered by the logic output. If 'on' the power from the POE (e-stop) will be connected, if 'off' the POE injector will be off.

The circuit diagram illustrates the main logic for the POE injector. It features a logic input connected to a 10k resistor (R12) to ground. The other end of R12 is connected to the base of a PNP transistor Q4. The emitter of Q4 is connected to VCC, and the collector is connected to the base of an NPN transistor Q3 (IRF9530). The emitter of Q3 is connected to VCC, and the collector is connected to the anode of a diode D1 (CGRM400-G). The cathode of D1 is connected to a 28k resistor (R13) to ground. The output of the circuit is labeled POE_INJO and is connected to the diode D1.



   	H1	H2	logicboard 2024-07-30 Sheet: 1/1
	M01MDJN0LH20E2.6	M02MDJN0LH20E2.6	
	H3	H4	
	M03MDJN0LH20E2.6	M04MDJN0LH20E2.6	

   	H1	H2	logicboard 2024-07-30 Sheet: 1/1
	M01MDUN0LH20E2.6	M02MDUN0LH20E2.6	
	H3	H4	
	M03MDUN0LH20E2.6	M04MDUN0LH20E2.6	

   	H1	H2	logicboard 2024-07-30 Sheet: 1/1
	M01MDJNOLBZ0E2.6	M02MDJNOLBZ0E2.6	
	H3	H4	
	M03MDJNOLBZ0E2.6	M04MDJNOLBZ0E2.6	